

Jiao, Yuchen

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Education and Employment

2025.7-present	Chinese University of Hong Kong	Research Associate
2025.3-2025.7	Chinese University of Hong Kong	Postdoc
2023.2-2025.2	Beijing Institute of Technology	Postdoc
2018.8-2023.1	Tsinghua University	Ph.D.
	Information and Communication Engineering	Advisor: Yuantao Gu
2014.8-2018.7	Tsinghua University	Bachelor of Engineering
	Electronic Engineering	Rank: 3/194

Research Interest

Generative AI (diffusion models), Reinforcement Learning, and Signal Processing.

Publications

* These authors contributed equally. Abbreviations: J (Journal), C (Conference), P (Preprint).

Generative AI

- P1. Instance-dependent Convergence Theory for Diffusion Models. 2026
Y. Jiao, and G. Li
minor revision at the *Journal of the Association for Computing Machinery (JACM)*. *arXiv:2410.13738*.
- P2. Transformers Meet In-Context Learning: A Universal Approximation Theory. 2026
G. Li*, Y. Jiao*, Y. Huang, Y. Wei, and Y. Chen
major revision at the *Annals of Statistics (AoS Invited Paper Session)*. *arXiv:2506.05200*.
- P3. Towards A Unified Framework for Guided Diffusion Models. 2026
Y. Jiao, Y. Chen, and G. Li.
reject and resubmit at the *Journal of the American Statistical Association (JASA)*. *arXiv:2512.04985*.
- C1. Are First-Order Diffusion Samplers Really Slower? A Fast Forward-Value Approach. 2026
Y. Jiao*, N. Li*, C. Cai, and G. Li.
International Conference on Machine Learning (ICML), Korea. *arXiv:2512.24927*
- C2. Provable Efficiency of Guidance in Diffusion Models for General Data Distribution. 2025
G. Li*, and Y. Jiao*.
International Conference on Machine Learning (ICML), Canada. *arXiv:2505.01382*
- C3. Improved Convergence Rate for Diffusion Probabilistic Models. 2025
G. Li*, and Y. Jiao*.
International Conference on Learning Representations (ICLR), **spotlight**, Singapore.
- P4. Optimal Convergence Analysis of DDPM for General Distributions. 2026
Y. Jiao, Y. Zhou, and G. Li.
submitted to Conference on Neural Information Processing Systems (NeurIPS), average score 5.0.
arXiv:2510.27562.

- P5. ConvergeFlow: Language Flow with Provable Convergence to Token Embeddings. 2026
N. Li*, Y. Jiao*, C. Cai, and G. Li.
arXiv:2608.23551.
- P6. Provable Diffusion-based Posterior Sampling for Linear Inverse Problems via DDIM. 2026
Y. Jiao*, N. Li*, C. Cai, Y. Chen, and G. Li.
arXiv:2607.19333.
- P7. Diffusion Models Adapt to Low-dimensional Structure under Flexible Coefficient Choices. 2026.
C. Cai*, Y. Jiao*, and G. Li.
arXiv:2606.23627.
- P8. Connections between Reinforcement Learning with Feedback, Test-time Scaling, and Diffusion Guidance: An Anthology. 2025
Y. Jiao, Y. Chen, and G. Li.
submitted to IEEE Transactions on Information Theory (IEEE TIT). *arXiv:2509.04372*.

Reinforcement learning

- P1. Provably Efficient Q-learning for Robust Reinforcement Learning. 2026
Y. Jiao, and G. Li.
minor revision at the *Journal of Machine Learning Research (JMLR)*.
- C1. Provable Memory Efficient Self-Play Algorithm for Model-free Reinforcement Learning. 2024 (corresponding author)
N. Li, Y. Jiao, H. Shan, S. Yan.
International Conference on Learning Representations (ICLR), Austria.
- P2. Sample Complexity of Average-Reward Q-Learning: From Single-Agent to Federated Reinforcement Learning. 2026
Y. Jiao*, J. Woo*, G. Li, G. Joshi, and Y. Chi.
submitted to Journal of Machine Learning Research (JMLR). *arXiv:2601.13642*.
- P3. Minimax-Optimal Multi-Agent Robust Reinforcement Learning.
Y. Jiao, and G. Li.
arXiv:2412.19873.

Signal Processing and Applications

- J1. Decentralized Rank-Adaptive Matrix Factorization – Part I: Algorithm Development. 2024
Y. Jiao, Y. Gu, T-H. Chang, and Z-Q Luo.
IEEE Transactions on Signal Processing (IEEE TSP), 73, 4124 - 4140.
- J2. Y. Jiao, Y. Gu, T-H. Chang, and Z-Q Luo. Decentralized Rank-Adaptive Matrix Factorization — Part II: Convergence Analysis. 2024
IEEE Transactions on Signal Processing (IEEE TSP), 73, 4141 - 4154.
- J3. Efficient Change-Point Detection over Fully Decentralized Wireless Networks with Low Communication Rate. 2024
Y. Jiao, L. Meng, Y. Li, Q. Yu, and Y. Gu.
IEEE Transactions on Vehicular Technology (IEEE TVT), 74(1):1626-1642.
- J4. Communication-Efficient Decentralized Subspace Estimation. 2022
Y. Jiao and Y. Gu.
IEEE Journal of Selected Topics in Signal Processing (IEEE J-STSP), 16(3):516-531.
- J5. Quantization for Communication-Efficient Change-Point Detection Over Networks. 2021
Y. Jiao, X. Xu and Y. Gu.
IEEE Transactions on Signal Processing (IEEE TSP), 70, 142-157.
- J6. Request-Response and Censoring-Based Energy-Efficient Decentralized Change-Point Detection With IoT Applications. 2021
Y. Gu, Y. Jiao, X. Xu and Q. Yu.
IEEE Internet of Things Journal (IEEE IoT), 8(8):6771-88.

- J7. Subspace Change-point Detection: A New Model and Solution. 2018
Y. Jiao, Y. Chen, and Y. Gu.
IEEE Journal of Selected Topics in Signal Processing (IEEE J-STSP), 12(6):1224-1239.
- C1. Incremental Aggregated Riemannian Gradient Method for Distributed PCA. 2023
X. Wang, Y. Jiao, H-T Wai, Y. Gu.
International Conference on Artificial Intelligence and Statistics (AISTATS), 7492-7510, Spain.
- C2. An Easy-to-Implement Framework of Fast Subspace Clustering For Big Data Sets. 2020
L. Meng, Y. Jiao, and Y. Gu.
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 3612-3616.
- C3. Principal Angle Detector For Subspace Signal With Structured Unknown Interference. 2020.
X. Xu, Y. Jiao, and Y. Gu.
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 6019-6023.
- C4. Pilot-Free Channel Change Detection for mmWave Massive MIMO System. 2019
Y. Wu, Y. Jiao, F. Gao, and Y. Gu.
IEEE Global Communications Conference (GLOBECOM), 1-6, USA.
- C5. An Efficient Algorithm for Hyperspectral Image Clustering. 2019
Y. Pan, Y. Jiao, T. Li, and Y. Gu.
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), UK.
- C6. Convergence Analysis on A Fast Iterative Phase Retrieval Algorithm without Independence Assumption. 2018
G. Li, Y. Jiao, and Y. Gu.
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Canada.
- C7. Principal Angles Preserving Property of Gaussian Random Projection for Subspaces. 2017
Y. Jiao, G. Li, and Y. Gu.
IEEE Global Conference on Signal and Information Processing (GlobalSIP), 318-322, Canada.

Grants

- G1. Feature-Enhanced Distributed Cooperative Detection Method for Underwater Acoustic Weak Targets, National Natural Science Foundation of China (NSFC)-Young Scientists Fund.
- G2. Distributed Cooperative Detection of Underwater Acoustic Targets under Low Signal-to-Noise Ratio and Narrow Bandwidth Conditions, China Postdoctoral Science Foundation-General Program.

Honors and Awards

Annals of Statistics Invited Paper Session, Joint Statistical Meetings	2026
Paper: "Transformers Meet In-Context Learning: A Universal Approximation Theory"	
Excellent graduate award, Tsinghua University	2018
Excellent graduate award, Beijing Municipal Education Commission	2018
Travel grant, GlobalSIP	2017
Meritorious Winner, Mathematical Contest In Modeling	2017

Scholarship and Fellowship

2023	Postdoctoral Fellowship Program.	China Postdoctoral Science Foundation
2022	Excellence Scholarship.	Tsinghua University
2020	Excellence Scholarship.	Tsinghua University
2018	Future Scholars Scholarship.	Tsinghua University

Teaching Experience

- T1. Signals and Systems, TA, Tsinghua EE, (2019 - 2022)
- T2. Matlab Programming, TA, Tsinghua EE, (2019 - 2022)